📘 50 Advanced MCQs

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Python Core & Advanced

1. What will be the output of the following?

```
def func(a, L=[]):
    L.append(a)
    return L

print(func(1))
print(func(2))
```

A. [1], [2]
B. [1], [1,2]
C. [1], [2,1]
D. Error

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2. What is the average time complexity of searching in a Python dictionary?
A. O(n)
B. O(log n)
C. O(1)
D. O(n log n)

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3. Which statement best describes Python decorators?
A. Modify variables globally
B. Wrap functions to extend behavior
C. Replace functions permanently
D. Only used in classes

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4. What does the expression `lambda x: x**2` produce?
A. Integer

B. Function object
C. List
D. Generator

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5. Which statement about recursion in Python is correct?
A. It does not use stack memory
B. It is always faster than loops
C. It uses call stack memory
D. It cannot return values

OOP Concepts

6. Which principle is primarily responsible for data hiding?
A. Abstraction
B. Encapsulation
C. Inheritance
D. Polymorphism

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7. Method overriding occurs when:
A. Same method exists in same class
B. Same method exists in parent and child class
C. Different method names are used
D. Static methods are used

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8. What does MRO stand for in Python?
A. Memory Runtime Object
B. Method Resolution Order
C. Module Runtime Operation
D. Multi Reference Object

Multithreading & Performance

9. Why is multithreading limited in Python?
A. Syntax limitations

B. Memory issues
C. Global Interpreter Lock
D. CPU architecture

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10. Multiprocessing is most suitable for:
A. I/O-bound tasks
B. CPU-bound tasks
C. UI rendering
D. Logging

NumPy & Pandas

11. What does `.iloc` in Pandas represent?
A. Label-based indexing
B. Integer-based indexing
C. Boolean indexing
D. Mixed indexing

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12. Which operation demonstrates vectorization?
A. Loop-based addition
B. NumPy array addition
C. File reading
D. String parsing

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13. What is NaN in data science?
A. String
B. Missing numeric value
C. Boolean
D. Integer

APIs & Backend

14. Why is FastAPI widely used?
A. Only synchronous support

B. No validation
C. Async support and performance
D. Requires Java

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15. What does HTTP status code 404 indicate?
A. Success
B. Unauthorized
C. Not Found
D. Server Error

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16. Which HTTP method is idempotent?
A. POST
B. PUT
C. PATCH
D. CONNECT

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17. REST stands for:
A. Random Execution System Transfer
B. Representational State Transfer
C. Resource Execution Style Transfer
D. Runtime Execution System Transfer

Databases

18. Which is not a SQL operation?
A. SELECT
B. INSERT
C. MAPREDUCE
D. DELETE

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19. MongoDB stores data as:
A. Tables
B. Rows
C. Documents
D. Graphs

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20. ORM is used for:
A. UI design
B. Database abstraction
C. Networking
D. Logging

DevOps

21. Docker is primarily used for:
A. Coding
B. Containerization
C. Testing
D. Debugging

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22. CI/CD stands for:
A. Continuous Integration / Continuous Deployment
B. Code Integration / Data Deployment
C. Continuous Inspection
D. None

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23. Which Git command stages changes?
A. git push
B. git commit
C. git add
D. git clone

Statistics

24. Mean is sensitive to:
A. Median
B. Outliers
C. Mode
D. Frequency

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25. Standard deviation measures:
A. Accuracy
B. Spread of data
C. Mean
D. Probability

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26. Probability values lie between:
A. -1 to 1
B. 0 to 1
C. 0 to 100
D. -∞ to ∞

Machine Learning

27. Linear regression is used for:
A. Classification
B. Regression
C. Clustering
D. Reinforcement

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28. Logistic regression is mainly used for:
A. Regression
B. Classification
C. Clustering
D. Dimensionality reduction

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29. Overfitting refers to:
A. Under learning
B. Model learning noise
C. Simple model
D. No training

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30. Train-test split helps in:
A. Increasing data
B. Preventing biased evaluation
C. Reducing features
D. Deployment

Clustering

31. KMeans requires:
A. Labels
B. Number of clusters
C. Supervision
D. Trees

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32. DBSCAN is useful for:
A. Linear data
B. Handling noise
C. Regression
D. Time series

Evaluation Metrics

33. Accuracy is unreliable in:
A. Balanced dataset
B. Imbalanced dataset
C. Small dataset
D. Large dataset

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34. Precision focuses on:
A. All predictions
B. Correct positive predictions
C. Negative predictions
D. Recall

Pipelines

35. Machine learning pipelines help in:
A. Visualization
B. Automating workflow
C. Data deletion
D. Model storage

Deep Learning

36. Activation functions introduce:
A. Linearity
B. Non-linearity
C. Errors
D. Data

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37. ReLU stands for:
A. Random Linear Unit
B. Rectified Linear Unit
C. Reduced Learning Unit
D. None

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38. CNN is best suited for:
A. Text
B. Images
C. Audio
D. Numbers

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39. Overfitting in deep learning can be reduced using:
A. Dropout
B. More layers
C. Less data
D. Higher learning rate

Advanced Concepts

40. Feature scaling means:
A. Data removal
B. Normalizing data
C. Encoding labels
D. Cleaning data

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41. Gradient descent is used for:
A. Data loading
B. Optimization
C. Visualization
D. Storage

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42. Learning rate controls:
A. Model size
B. Step size
C. Dataset size
D. Accuracy

Scenario-Based

43. Which model is suitable for email spam detection?
A. Linear regression
B. Logistic regression
C. KMeans
D. PCA

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44. Best algorithm for customer segmentation:
A. Logistic regression
B. KMeans
C. Linear regression
D. CNN

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45. High training accuracy but low test accuracy indicates:
A. Underfitting
B. Overfitting
C. Balanced model
D. Optimized model

System Design

46. Which tool is used for API testing?
A. Docker
B. Postman
C. Git
D. NumPy

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47. JSON is:
A. Database
B. Data format
C. Programming language
D. Protocol

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48. FastAPI uses which library for validation?
A. Flask
B. Pydantic
C. Django
D. SQL

Practical ML

49. Which technique improves generalization?
A. Noise addition
B. Regularization
C. Overfitting
D. Random weights

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50. Best way to deploy ML models in production:
A. Local script
B. REST API
C. Excel
D. Text file